

Folded-Cascode OTA Sizing Preparation

<https://gemini.google.com/app/f8b0e76a25d43d75>

User prompt: how to do that

Response: To execute this transient stability and settling time test, you need to configure your input sources, set the simulation parameters, and analyze the resulting waveforms.

Here is the exact step-by-step procedure in Virtuoso:

1. Configure the Differential Input Pulse To create a 100mV *differential* pulse from an initial state of 0, your positive input must step up by 50mV, and your negative input must step down by 50mV. Open the properties (q) of the two `vpulse` sources at the front of your testbench.

- **Positive `vpulse`:**

- Voltage 1: 0
- Voltage 2: 50m
- Delay time: 1u
- Rise time: 10n
- Fall time: 10n
- Pulse width: 1u
- Period: 2u

- **Negative `vpulse`:**

- Voltage 1: 0
- Voltage 2: -50m
- (All time parameters remain exactly the same).

2. Setup the Transient Analysis

- In ADE, go to **Analyses** → **Choose** → **tran**.
- Set Stop Time to 3u.
- Click **Options** at the bottom of the window, locate the **maxstep** field, and enter 10n.

3. Plot the Signals

- Run the simulation.
- Plot the transient voltages (VT) for VINP, VINN, VOUTP, and VOUTN.

- To plot the output common-mode voltage (VOCM), use the Calculator to average the outputs: $(VT("/VOUTP") + VT("/VOUTN")) / 2$, and append it to your plot.

4. Check for Ringing (Stability Analysis)

- **Differential Loop:** Zoom in on the output transition right after $1\mu s$. If the output curves smoothly into its final value, your phase margin is excellent. If it overshoots and bounces (rings) multiple times, your differential phase margin is too low.
- **CMFB Loop:** Look at your VOCM trace. It should stay locked at your 1.25V reference. You may see small, sharp spikes exactly at the $1\mu s$ and $2\mu s$ switching edges, but they should snap right back to 1.25V without prolonged wiggling.

5. Calculate the 1% Settling Time Because your closed-loop gain is 2, a 100mV differential input step forces a 200mV differential output step.

- 1% of a 200mV step is **2mV**.
- Your output must settle within a tight band of **198mV to 202mV**.
- Find the exact timestamp where the differential output ($VOUTP - VOUTN$) permanently crosses into that $\pm 2mV$ band and subtract the $1\mu s$ start time.
- *Cadence Shortcut:* You can open the Calculator and use the `settlingTime` function on your differential output trace, setting the initial value to 0, final value to 200m, and error band to 1%.

User prompt: Calculate the 1% Settling Time how

Response: To find the exact 1% settling time, you need to measure the time delay between the start of your input pulse and the moment your output permanently locks into its final error band.

Because your closed-loop gain is 2, your 100mV differential input step will force a 200mV differential output step. 1% of 200mV is 2mV. This means your signal is considered "settled" the moment it enters the 198mV to 202mV range and never leaves it.

Method 1: The Manual Graph Method

1. Plot your differential output by evaluating $VT("/VOUTP") - VT("/VOUTN")$.
2. Press **v** on your keyboard to place a vertical marker at exactly $1\mu s$ (the moment your input pulse starts).
3. Press **h** to place two horizontal markers: one at 198mV and one at 202mV.
4. Zoom in heavily on the curve where the signal rises into this band.
5. If your signal rings (overshoots and bounces), find the exact timestamp where the waveform crosses into that 198mV–202mV tunnel for the final time and does not bounce back out. Place a second vertical marker there.

6. The time difference (ΔX) between your $1\mu\text{s}$ start marker and this final crossing marker is your 1% settling time.

Method 2: The Cadence Calculator Shortcut

1. Open the Virtuoso Calculator and select the **settlingTime** function from the function panel.
2. **Signal:** Enter your differential expression: $VT("/VOUTP") - VT("/VOUTN")$.
3. **Initial Value:** Type 0.
4. **Final Value:** Type 200m .
5. **Percentage:** Type 1.
6. Click the evaluate button (the gear icon). The calculator will instantly output the exact settling time in seconds.

Compare this measured value against your lab manual's target specification (e.g., 100ns). If it is too slow, you will need to increase the GBW by pushing more bias current into your main OTA tail.

User prompt: how to do the manual graph

Response: Here is exactly how to measure the settling time visually using the markers in the Virtuoso Visualization and Analysis (ViVA) graph window.

1. Plot the Differential Signal

- Open your Calculator.
- Select vt (transient voltage) and click your VOUTP wire in the schematic.
- Click the - (minus) button in the Calculator.
- Select vt again and click your VOUTN wire.
- Click the **Plot** icon (the graph with a red jagged line) in the Calculator to plot the expression.

2. Create the 1% Error Band (Horizontal Markers)

- Click anywhere inside the graph window to make it active.
- Press the **h** key on your keyboard. A horizontal marker will appear attached to your cursor.
- Click to place it, then double-click the marker's label. Change its Y-value manually to 198m .
- Press **h** again, place a second marker, and change its Y-value to 202m .

- You have now created the visual $\pm 2\text{mV}$ tunnel that your signal must settle into.

3. Mark the Start Time (Point A)

- Press the **a** key on your keyboard. This places the first half of a "Delta Marker."
- Double-click the marker it drops, and manually set its X-value (time) to exactly 1u (the moment your input pulse started).

4. Find the Settling Point (Point B)

- Zoom in heavily (click and drag a box) around the area where your signal rises and crosses into your horizontal error band.
- Trace the signal with your eyes. If it overshoots the 202m line and bounces back down, follow it until it crosses back inside.
- Find the exact timestamp where the line crosses into the 198m–202m band for the *absolute final time* and stays flat.
- Press the **b** key on your keyboard and click exactly on that final crossing point.

5. Read the Result

- Because you used the **a** and **b** keys, Cadence automatically calculates the difference between the two points.
- A small text box will appear on the screen connecting Point A and Point B. Look for the **dx** value (delta X).
- That dx time is your exact 1% settling time.

User prompt: how to fix the settling time if its bigger than 100n

Response: If your settling time is failing the 100ns specification, your amplifier is either starved for bandwidth (rising too slowly) or suffering from poor phase margin (ringing and bouncing out of the error band).

Look closely at the shape of your transient output curve and apply the corresponding fix:

Scenario A: The signal rises smoothly, but it takes too long to reach 198mV. This means your closed-loop Gain-Bandwidth Product (GBW) or your Slew Rate is too low. You must increase the transconductance (g_m) of your input pair.

- **Increase the Tail Current (Most Effective):** Pump more current into your main OTA tail transistor. If your tail is currently $20\mu\text{A}$, increase it to $30\mu\text{A}$ or $40\mu\text{A}$. You will also need to re-extract the widths of your input pair to maintain their g_m/I_D operating point. This simultaneously boosts your small-signal GBW and your large-signal slew rate.

- **Increase Input Pair Width ($\bar{W}$):** If you cannot afford more power consumption, increase the width of your input transistors (M0 and M1). This pushes them deeper into weak inversion, extracting more g_m for the exact same amount of bias current.

Scenario B: The signal shoots up fast, but overshoots 202mV and rings up and down. This means your phase margin is too low. The non-dominant pole at your folding node is too close to your Unity Gain Frequency, causing the amplifier to become underdamped.

- **Increase Cascode Branch Current:** The non-dominant pole is determined by the capacitance at the folding node and the resistance looking into the source of your cascode transistors ($1/g_{m,casc}$). By shifting some of your current budget to increase the bias current flowing through the side branches (the NMOS sinks and PMOS sources), you increase the cascode g_m , which pushes the non-dominant pole to a much higher frequency and stabilizes the ringing.
- **Shrink Transistor Widths at the Folding Node:** Parasitic capacitance at the folding node drags down the phase margin. If your top PMOS current sources (M3, M4) or your input pair (M0, M1) are extremely wide, reduce their widths to slash their parasitic drain capacitance.

Make one adjustment at a time, check your DC operating points to ensure no transistors fell into the triode region, and re-run the transient simulation.

User prompt: this is cm pulse what is this appeared ?

Response: The spikes you are seeing at exactly 1.0 μ s and 2.0 μ s are the **Common-Mode Feedback (CMFB) loop actively regulating the amplifier.**

Here is the step-by-step breakdown of the physics happening in that plot:

1. The Instantaneous Spike (Capacitive Feedthrough) At 1.0 μ s, your input common-mode voltage suddenly steps up by 100mV. Because your amplifier is wrapped in a capacitive feedback network (C_s and C_f), these capacitors act as short circuits to instantaneous, high-frequency voltage changes. The 100mV jump at the input couples directly through the capacitors, causing your output common-mode voltage to instantly spike upward.

2. The Smooth Recovery (CMFB Action) The moment the output voltage spikes, it deviates from your desired DC reference voltage. Your CMFB error amplifier instantly senses this error. It adjusts the control voltage (VCTRLP) on the OTA's tail transistor, altering the current to actively pull the output voltage back down to its designated resting level.

3. Loop Stability The fact that the output smoothly curves back to its baseline—without bouncing up and down (ringing) multiple times—proves that your CMFB loop is **highly stable** with an excellent phase margin. If your CM loop had a poor phase margin, that spike would have triggered a long, oscillating wave.

One Important Observation: Look at the Y-axis value where your red/yellow/green traces settle. They are resting at roughly **1.06V**. If your lab specification requires an output common-mode voltage of exactly **1.25V**, your CMFB circuit has a systematic DC offset. This is usually caused by either:

- Low DC gain in the CMFB error amplifier.
- A mismatch between the V_{CM} Common-Drain buffers and the V_{REF} dummy buffer you built earlier.

The transient dynamic behavior is perfectly correct, but you may want to check your CMFB transistor sizing if you need that baseline to sit exactly at 1.25V.

*User prompt: I am writing an Analog IC Design lab report and need your help to draft it step-by-step. I will provide you with the following materials: 1. The current lab manual/instructions. 2. Reference data or a report from a previous lab (if we need to pull older numbers). 3. A template document for the formatting. 4. Images of my simulation plots and schematics (e.g., AC sweeps, transient analysis, STB analysis). Please act as an expert analog electronics design assistant. Follow these strict rules while helping me write the report: **1. Step-by-Step Execution:** * Start by generating a comprehensive, well-structured outline based on the lab manual's deliverables. * Once I approve the outline, we will write the report **one section at a time**. * **DO NOT** move to the next section or point without my explicit confirmation. **2. Data Extraction & Hand Calculations:** * Extract the relevant simulation values directly from the images/data I provide. * Perform the necessary theoretical hand calculations (e.g., feedback factor, closed-loop gain, bandwidth, pole locations) using the data from previous labs and standard analog design formulas. * Present comparisons between Hand Calculations and Simulation Results in clean, easy-to-read tables. **3. Insightful Technical Comments:** * Provide deep, accurate engineering commentary for each section. * Explain the *why* behind the results (e.g., explaining gain desensitization, capacitive loading effects on GBW, systematic offsets, or finite open-loop gain errors). **4. Formatting for MS Word:** * Keep the formatting clean and structured using standard Markdown (Headings, bold text, bullet points). * Avoid complex LaTeX formatting unless it is for a specific math equation, as the final text will be copy-pasted directly into Microsoft Word. walk part by part and write the captions of pictures i will give to u*

Response: Please upload or paste the text of the current lab manual and the list of required deliverables. Once you provide those details, I will generate a comprehensive, well-structured outline for your approval before we write the first section.

Do you have a specific template document you want to use, or should I structure the outline based on standard academic lab report formats?

Response: I have received the lab manual and the g_m/I_D chart images you provided. Based on the "Report the following" directives in the lab manual, I have generated a comprehensive outline for your lab report.

Please review the proposed structure below.

Proposed Lab Report Outline

1. Introduction & Design Specifications

- Brief overview of the fully-differential folded-cascode OTA and the GF180MCU technology targets.
- Summary table of the target specifications (Gain, GBW, Phase Margin, Settling Time, etc.).

2. Part 1: g_m/I_D Design Charts

- Presentation of the generated charts for NMOS and PMOS (V_A , W/I_D , f_T , V_{GS} vs. g_m/I_D).
- Brief technical commentary on the trade-offs observed in the charts (e.g., intrinsic gain vs. transit frequency).

3. Part 2 & 3: OTA Design and Open-Loop Simulation (Behavioral CMFB)

- **3.1 DC Operating Point:** Schematic with annotated DC voltages/currents. Analysis of the output common-mode (CM) level and the ideal error amplifier voltages.
- **3.2 AC Analysis:** Differential small-signal plots (Gain and Phase vs. Frequency).
- **3.3 Performance Table:** Comparison of simulated parameters (A_0 , BW , GBW , UGF , PM) versus theoretical hand calculations. Technical commentary on any deviations.

4. Part 4: Open-Loop OTA Simulation (Actual CMFB)

- **4.1 Transistor-Level CMFB Design:** Schematic with annotated DC OP. Commentary on the CM level at the OTA output and the error amplifier input/output relationship (including the V_{GS} level-shift dummy buffer).
- **4.2 AC Analysis:** Differential small-signal plots and updated performance table (A_0 , BW , GBW , UGF , PM) with the real CMFB loading.

5. Part 5: Closed-Loop Simulation (AC and STB Analysis)

- **5.1 DC Operating Point:** Schematic of the closed-loop configuration. Analysis of the CM levels at both the OTA input and output.
- **5.2 Differential Closed-Loop Response:** AC plot of V_{ODIFF} . Extraction of DC closed-loop gain, CL BW, and CL GBW.
- **5.3 Stability Analysis (STB):** Overlaid Bode plots (Loop Gain and Phase) for both Differential and CM loops. Technical comparison of GBW and PM between the two loops, and a comparison of the STB loop parameters versus the open-loop AC parameters (explaining the β factor and loading effects).

6. Part 6: Closed-Loop Transient Analysis

- **6.1 Differential Input Pulse:** Transient plots (V_{INP} , V_{INN} , V_{OUTP} , V_{OUTN} , V_{OCM}). Analysis of ringing/stability. Calculation and verification of the 1% settling time.

- **6.2 Common-Mode Input Pulse:** Transient plots evaluating the stability of the CMFB loop to CM steps.
- **6.3 Output Swing:** Transient plots for a 150mV amplitude 100kHz sinusoidal input. Calculation of differential input/output peak-to-peak swings and verification of the closed-loop linear gain.

7. Conclusion

- Brief summary of the design outcomes, trade-offs made to meet the 100ns settling time, and final performance against specs.

Do you approve of this outline? If yes, please confirm, and we will begin writing **Section 1 and Section 2** using the chart images you have already provided. Remember, we will proceed strictly one section at a time.

User prompt: ok nice also write all hand analysis and answer all comments and if anything u need tell me what u need

Response: 1. Introduction & Design Specifications

This report details the design and simulation of a fully-differential folded-cascode Operational Transconductance Amplifier (OTA) with common-mode feedback (CMFB). The circuit is implemented using the GF180MCU technology node. The primary objective is to balance trade-offs between open-loop gain, bandwidth, phase margin, and settling time utilizing the g_m/I_D design methodology.

The target specifications for the amplifier are summarized in the table below:

Parameter	Specification (GF180MCU)
Supply Voltage (V_{DD})	2.5 V
Closed Loop Gain (A_{CL})	2 V/V
Phase Margin (at target A_{CL})	$\geq 70^\circ$
CM Input Range - Low	≤ 0 V
CM Input Range - High	≥ 1 V
Differential Output Swing	1.2 V peak-to-peak
Load Capacitance (C_L)	500 fF
DC Loop Gain	60 dB

Parameter	Specification (GF180MCU)
Closed-Loop Settling Time (1% error)	100 ns

2. Part 1: g_m/I_D Design Charts

To size the transistors efficiently without relying on square-law approximations, lookup table (LUT) charts were generated for both NMOS (nch_33) and PMOS (pch_33) devices across varied channel lengths ($L = 280\text{nm}$ to $5\mu\text{m}$).

(Insert nch_VA.png here) **Figure 2.1:** NMOS Early Voltage (V_A) vs. g_m/I_D .

(Insert pch_VA.png here) **Figure 2.2:** PMOS Early Voltage (V_A) vs. g_m/I_D .

(Insert nch_ft.png here) **Figure 2.3:** NMOS Transit Frequency (f_T) vs. g_m/I_D .

(Insert pch_ft.png here) **Figure 2.4:** PMOS Transit Frequency (f_T) vs. g_m/I_D .

(Insert nch_W_ID.png here) **Figure 2.5:** NMOS W/I_D vs. g_m/I_D .

(Insert pch_W_ID.png here) **Figure 2.6:** PMOS W/I_D vs. g_m/I_D .

(Insert nch_vgs.png here) **Figure 2.7:** NMOS V_{GS} vs. g_m/I_D .

(Insert pch_vgs.png here) **Figure 2.8:** PMOS V_{GS} vs. g_m/I_D .

Technical Commentary on Device Sizing Trade-offs

The generated charts illustrate the fundamental trade-offs in analog IC design that govern the sizing of the folded-cascode OTA:

- Intrinsic Gain vs. Speed:** Figures 2.1 and 2.2 demonstrate that the Early Voltage (V_A , which dictates intrinsic gain $g_m r_o$) increases significantly as channel length (L) increases. However, Figures 2.3 and 2.4 show that increasing L drastically reduces the transit frequency (f_T), penalizing the device's speed. Consequently, minimum channel lengths (280nm) are reserved for the input differential pair to maximize Gain-Bandwidth Product (GBW), while longer lengths (0.5 μm to 1.0 μm) are utilized for the cascode and current source transistors to achieve the 60 dB loop gain target.
- Efficiency vs. Area:** Figures 2.5 and 2.6 show that operating in weak inversion (high g_m/I_D) yields maximum transconductance per unit of bias current. However, the W/I_D ratio increases exponentially in this region, requiring massive device widths. The input pair is biased in moderate/weak inversion (e.g., $g_m/I_D = 15$ to 18) to optimize GBW without

excessive parasitic capacitance loading, while the current mirrors are pushed deeper into strong inversion (e.g., $g_m/I_D = 8$ to 10) to minimize area and improve matching.

- **Voltage Headroom:** Figures 2.7 and 2.8 confirm that lower g_m/I_D values demand higher V_{GS} (and correspondingly higher overdrive voltage, V_{DSAT}). To satisfy the 1.2V peak-to-peak output swing and the wide input common-mode range, the cascode stack must be carefully biased to ensure all devices remain in saturation under maximum signal excursion.

Please provide the schematic image with the DC operating point annotations and the AC small-signal plots (Gain and Phase vs. Frequency) for **Section 3 (Open-Loop OTA with Behavioral CMFB)**. If you have the exact g_m , I_D , and r_o values from your simulation, please provide them so I can type out the step-by-step theoretical hand calculations for the DC Gain and GBW.

User prompt: Lets do sizing and hand analysis of sizing first based on the specs required M1 is input pair M2 is nmos sink sources m3 is Cascode nmos M4 is the cascode pmos M5 is the pmos current mirrors M6 is the tail pmos current source

Response: 3. Part 2 & 3: OTA Sizing and Hand Analysis

Based on the LUT extraction images provided, here is the complete sizing breakdown and the theoretical hand analysis for your fully-differential folded-cascode OTA.

3.1 Transistor Sizing and Biasing Extraction

The extracted values follow a textbook g_m/I_D methodology for a folded-cascode architecture, balancing the current split (10 μ A per branch) and optimizing inversion regions for speed versus noise.

Device	Function	I_D (μ A)	g_m/I_D	L (μ m)	W (μ m)	g_m (μ S)
M1	Input Pair (PMOS)	10	16.5	0.3	20.88	165
M2	Sink Mirror (NMOS)	20	8.0	1.0	5.52	160
M3	Cascode (NMOS)	10	9.0	1.0	3.56	90
M4	Cascode (PMOS)	10	9.0	1.0	15.32	90
M5	Current Source (PMOS)	10	7.0	1.0	8.84	70
M6	Tail Source (PMOS)	20	7.0	1.0	17.68	140

3.2 Sizing Justification & Technical Commentary

- **Input Pair (M1):** Biased in moderate/weak inversion ($g_m/I_D = 16.5$) to maximize transconductance for a given current, which directly dictates the Gain-Bandwidth Product (GBW). A short channel length ($L = 0.3\mu\text{m}$) is chosen to minimize parasitic gate and drain capacitances, keeping the non-dominant poles at high frequencies.
- **Current Mirrors & Sinks (M2, M5, M6):** These devices contribute significantly to the OTA's total noise and offset. To minimize their noise contribution (which is proportional to their g_m), they are pushed into strong inversion ($g_m/I_D = 7$ to 8). A longer channel length ($L = 1.0\mu\text{m}$) is used to maximize their output resistance (r_o) and improve current matching.
- **Cascode Devices (M3, M4):** These transistors do not contribute significantly to low-frequency noise, so their g_m does not need to be minimized. They are biased at $g_m/I_D = 9$ with $L = 1.0\mu\text{m}$. This provides enough transconductance to maintain a high-frequency pole at the folding node ($\approx g_{m3}/C_{fold}$), while the $1.0\mu\text{m}$ length ensures a massive intrinsic gain ($g_m r_o$) to boost the final DC output impedance.
- **Current Split:** The tail current is $20\mu\text{A}$, supplying $10\mu\text{A}$ to each M1 transistor. The bottom NMOS sink (M2) pulls $20\mu\text{A}$. Therefore, $10\mu\text{A}$ comes from the input pair, and the remaining $10\mu\text{A}$ is pulled through the cascode stack, achieving a perfectly balanced 1:1 current split.

3.3 Theoretical Hand Calculations

Using the extracted g_m values and the lab manual's 500 fF load capacitance, we can theoretically predict the amplifier's AC and transient performance.

1. Open-Loop Gain-Bandwidth Product (GBW): In a single-stage folded-cascode OTA, the GBW is dictated entirely by the input pair's transconductance and the load capacitance.

$$GBW_{OL} = \frac{g_{m1}}{2\pi \cdot C_L}$$

$$GBW_{OL} = \frac{165\mu\text{S}}{2\pi \cdot 500\text{ fF}} \approx \mathbf{52.5\text{ MHz}}$$

2. Slew Rate (SR): The maximum rate of change at the output is limited by the total available current to charge the load capacitor.

$$SR = \frac{I_{tail}}{C_L}$$

$$SR = \frac{20\mu\text{A}}{500\text{ fF}} = \mathbf{40\text{ V}/\mu\text{s}}$$

3. DC Open-Loop Gain (A_0): The DC gain is the product of the input transconductance and the total output impedance (R_{out}).

$$A_0 = g_{m1} \cdot R_{out}$$

$$R_{out} = R_{up} \parallel R_{down}$$

Where:

- $R_{up} \approx g_{m4} \cdot r_{o4} \cdot r_{o5}$
- $R_{down} \approx g_{m3} \cdot r_{o3} \cdot (r_{o1} \parallel r_{o2})$

(Note: While exact r_o values require the V_A or g_{ds} outputs from the LUT, standard $1\mu\text{m}$ devices in $130\text{nm}/180\text{nm}$ nodes typically exhibit intrinsic gains ($g_m r_o$) of 20 to 40 V/V. Given the double-cascode structure, R_{out} will easily be in the tens of Mega-ohms, yielding an $A_0 = G_m R_{out}$ comfortably exceeding the 60 dB (1000 V/V) specification).

User prompt: write the hand analysis of peak to peak output

Response: Here is the theoretical hand analysis for the peak-to-peak output swing to add to your report.

3.4 Hand Analysis of Peak-to-Peak Output Swing

To determine the maximum theoretical output swing, we must first estimate the overdrive voltage (V_{DSAT} or V^*) required to keep each transistor in the output branch (M2, M3, M4, M5) safely in the saturation region.

Using the standard approximation $V_{DSAT} \approx \frac{2}{g_m/I_D}$, we can extract the minimum voltage drops across the output stack based on our chosen sizing:

- **M5 (PMOS Source):** $g_m/I_D = 7 \implies V_{DSAT5} \approx \frac{2}{7} \approx 285 \text{ mV}$
- **M4 (PMOS Cascode):** $g_m/I_D = 9 \implies V_{DSAT4} \approx \frac{2}{9} \approx 222 \text{ mV}$
- **M3 (NMOS Cascode):** $g_m/I_D = 9 \implies V_{DSAT3} \approx \frac{2}{9} \approx 222 \text{ mV}$
- **M2 (NMOS Sink):** $g_m/I_D = 8 \implies V_{DSAT2} \approx \frac{2}{8} \approx 250 \text{ mV}$

1. Raw OTA Output Swing (Without CMFB Loading) The absolute maximum and minimum single-ended voltages at the folding node (before pushing transistors into the triode region) are:

$$V_{out,max} = V_{DD} - |V_{DSAT5}| - |V_{DSAT4}| = 2.5 \text{ V} - 0.285 \text{ V} - 0.222 \text{ V} = \mathbf{1.993 \text{ V}}$$

$$V_{out,min} = V_{DSAT2} + V_{DSAT3} = 0.250 \text{ V} + 0.222 \text{ V} = \mathbf{0.472 \text{ V}}$$

The raw single-ended peak-to-peak swing is $1.993 \text{ V} - 0.472 \text{ V} = 1.521 \text{ V}$. Because this is a fully differential amplifier, the raw differential output swing is simply twice the single-ended swing (3.042 V_{pk-pk}).

2. CMFB-Limited Output Swing As noted in the lab instructions, attaching the actual CMFB circuit introduces a stricter limit on the positive output swing. The PMOS Common-Drain (CD) buffers used to sense the output voltage without loading the OTA require a gate-to-source voltage ($|V_{GSP}|$) drop.

- The lab manual defines the new CMFB-limited maximum output voltage as $V_{DD} - V^* - |V_{GSP}|$.
- Assuming the CD buffer requires a typical $|V_{GSP}| \approx 0.95 \text{ V}$ and its current source requires $V^* \approx 0.25 \text{ V}$, the positive swing is clipped at:

$$V_{out,max(CMFB)} = 2.5 \text{ V} - 0.25 \text{ V} - 0.95 \text{ V} \approx \mathbf{1.30 \text{ V}}$$

With the CMFB buffer attached, the valid single-ended output range shrinks to **0.472 V to 1.30 V**.

- Final Single-Ended Swing:** $1.30 \text{ V} - 0.472 \text{ V} = \mathbf{0.828 \text{ V}}$
- Final Differential Swing:** $2 \times 0.828 \text{ V} = \mathbf{1.656 \text{ V}_{pk-pk}}$

Conclusion: The theoretical CMFB-limited differential swing of 1.656 V_{pk-pk} successfully exceeds the GF180MCU target specification of $\geq 1.2 \text{ V}_{pk-pk}$. To maximize this symmetric swing in the transient simulation, the output common-mode level (V_{OCM}) should be placed exactly in the middle of this valid range ($\approx 0.88 \text{ V}$).

User prompt: make a table with final sizing we got from part 2

Response:

Transistor	Function	Type	$I_D (\mu\text{A})$	g_m/I_D	$L (\mu\text{m})$	$W (\mu\text{m})$
M1	Input Pair	PMOS	10	16.5	0.3	20.88
M2	Sink Mirror	NMOS	20	8.0	1.0	5.52
M3	Cascode	NMOS	10	9.0	1.0	3.56
M4	Cascode	PMOS	10	9.0	1.0	15.32
M5	Current Source	PMOS	10	7.0	1.0	8.84
M6	Tail Source	PMOS	20	7.0	1.0	17.68